Predicting 48-Hour Closure of NYC Street-Condition Requests

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GitHub Repository

Introduction

Our group made the decision to research the length of time it takes to close street-condition requests in New York City, including complaints about potholes, cave-ins, and unsuccessful repairs. We wanted to find out if personnel might prioritize follow-up and provide residents with more reasonable expectations for closing times by identifying requests that are likely to take longer. One earlier study that is relevant to our subject is SWIFT [1], which forecasts response times for the upcoming week based on data from the preceding two weeks. It models correlations between past and future response times using Gaussian Conditional Random Fields. The authors reported reduced prediction errors when comparing their method to time-series forecasting and linear regression. Our experiment examines whether information at the time of submission can predict administrative closure within 48 hours for NYC DOT street-condition requests issued in 2025.

Data

The "311 Service Requests from 2020 to Present" dataset [2] from NYC Open Data provided our data. By filtering for the DOT agency, the "Street Condition" complaint type, and requests made in 2025, we gathered records via its API. Month by month, we gathered the data, and when necessary, we used pagination to acquire subsequent batches of up to 5,000 items. We checked for duplicate request IDs and missing information, converted the submission and closure timestamps to datetime values, and calculated the time difference in hours. For the following months, we automated collection and cleaning by utilizing reusable functions. The original records were retained as JSON files, and the processed monthly datasets were stored as CSV files before being integrated into a single table.

We collected 70,330 requests and discovered no duplicate IDs. We excluded six records because their closing dates came before their creation dates. Of the remaining records, 3,475 requests had no closure date and were listed as Pending or Open. We kept them and categorized them as not closed within 48 hours, presuming that the recorded statuses accurately reflected non-closure at collection and that sufficient follow-up time had elapsed. Two requests with missing closure dates and unspecified statuses were omitted because the outcomes were uncertain. This left 70,322 labeled requests.

The final labeled dataset has 70,322 requests and 38 columns. The request IDs are still kept in the saved data for tracking and duplicate checking but will be removed from the predictor matrix.

Exploratory Data Analysis

Our exploratory data analysis inspected data quality, feature characteristics, and associations between our features and recorded closure within 48 hours. We derived submission hour, month, weekday, and weekend status from each request's `created_date` timestamp. We also aggregated counts of earlier requests citywide over the previous 24 hours, within the same borough over the previous seven days, and within the same problem category over the previous seven days. The first week of January was excluded from history-count comparisons because the preceding December records were unavailable.

We checked the 70,322 labeled requests for duplicate identifiers, missing values, and unusual closure durations. No duplicate request IDs were found. However, 19,601 requests (27.9%) lacked latitude and longitude, almost all associated with "BLOCKFACE" addresses. We retained these observations and created a missing-location indicator because removing them would disproportionately exclude this address category. Another 3,475 requests lacked closure dates and were marked Pending or Open. We also identified 3,276 zero-hour closures, all associated with the UNKNOWN submission channel, which warrant further investigation.

Across the labeled dataset, 63.1% of requests were recorded as closed within 48 hours, while 36.9% were labeled otherwise, including unresolved requests. This establishes a 63.1% majority-class accuracy baseline. A histogram with logarithmically spaced bins and a logarithmic horizontal axis showed the wide range of closure durations. Among requests with positive recorded durations, the median was 27.6 hours and the mean was 72.7 hours, indicating a long right tail. A reference line marked the 48-hour threshold.

Bar charts showed that Queens and Brooklyn together contributed 63.1% of requests, while potholes accounted for 54.6%. Normalized stacked bars compared closure proportions across problem types, boroughs, submission channels, and calendar features. Using proportions allowed comparisons between groups of different sizes. Potholes had a 74.5% quick-closure rate, compared with 44.3% for cave-ins and 42.9% for failed street repairs, suggesting that problem type may be useful for prediction.

Three explanatory figures highlight the main findings. Figure 1 shows that 74.0% of Tuesday requests closed within 48 hours, compared with 29.2% on Friday. Only 3.8% of recorded closures occurred on weekends, suggesting that the placement of the 48-hour window matters. This supports including submission-day features when identifying requests likely to remain open longer.

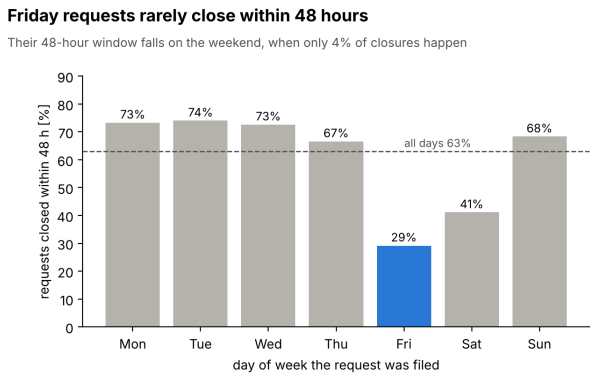


Figure 1: Percentage of requests recorded as closed within 48 hours, grouped by submission weekday.

Figure 2 shows a 39.0% quick-closure rate in the Bronx, compared with approximately 61–72% in other boroughs. Horizontal bars and sample-size labels make these geographic comparisons clear, although differences in request composition may contribute to the observed gaps. These comparisons identify areas deserving further investigation and motivate evaluating whether the model performs consistently across boroughs.

Figure 3 presents monthly request volumes and closure proportions in separate panels. Quick closures declined from 77.7% in January to 54.7% in July before fluctuating later in the year. This reveals variation that should inform model evaluation, even without using a time-series model. Additional density histograms showed substantial

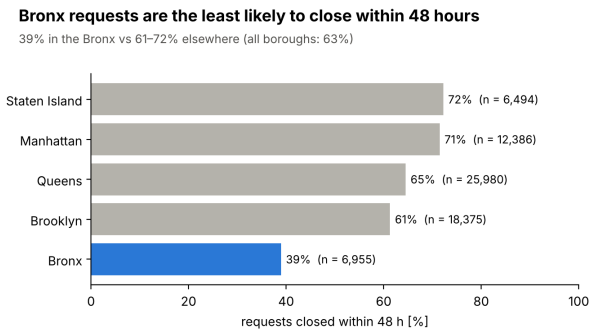


Figure 2: Percentage of requests recorded as closed within 48 hours by borough, with sample sizes shown for context.

overlap between outcomes for citywide and borough request volumes, so their predictive value remains uncertain.

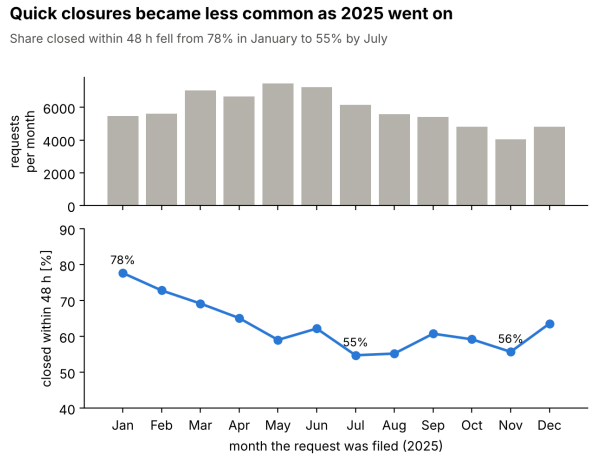


Figure 3: Monthly request volumes and percentages recorded as closed within 48 hours during 2025, displayed in separate panels.

Together, our findings connect the binary classification task to our ultimate goal of helping NYC officials better recognize requests that may require more attention or additional follow-up. Differences across problem types, boroughs, and submission dates, and their associations with the 48-hour closure rate, suggest that a single overall rate may not provide accurate expectations for individual requests. A model combining these characteristics could provide more useful estimates of administrative turnaround.

Problem Description and Potential Impact

Our project focuses on Street Condition service requests assigned to the New York City Department of Transportation, including potholes, cave-ins, defective hardware, and failed street repairs. Using the NYC Open Data 311 Service Requests from 2020 to Present dataset [2], we assembled a labeled subset of 70,322 requests filed during 2025. Each observation represents one service request.

We plan to train a supervised binary classification model to predict, at the time of filing, whether a request will be recorded as closed within 48 hours. The target is constructed from the difference between creation and closure timestamps, with unresolved Pending or Open requests assigned to the negative class under the assumption of sufficient follow-up. Predictors will be limited to information available at submission, such as problem type, location, submission channel, and calendar characteristics. Closure dates, current status, and resolution descriptions will be excluded from model inputs to prevent information leakage.

A useful model could help officials identify requests likely to remain open longer, support decisions about follow-up, and provide more realistic expectations about administrative turnaround. Examining performance across boroughs and problem types could also identify patterns warranting further operational investigation.

These predictions would measure recorded closure rather than verified repair. In our dataset, 3,397 closed records explicitly identify duplicate complaints whose original requests were still being addressed. Other resolution descriptions document referrals or further inspections, and NYC311 explains that DOT refers cave-in repairs to responsible agencies or utilities. Therefore, the model would support human review rather than independently determine service priority. Geographic differences alone would not establish neglect or its causes, and the 48-hour threshold is an analytical choice rather than a universal

DOT service deadline.

References

[1] R. Raj, A. Ramesh, A. Seetharam, and D. DeFazio, "SWIFT: A non-emergency response prediction system using sparse Gaussian Conditional Random Fields," *Pervasive and Mobile Computing*, vol. 71, article 101317, 2021. https://doi.org/10.1016/j.pmcj.2020.101317

[2] NYC Open Data, "311 Service Requests from 2020 to Present." [Online]. Available: https://data.cityofnewyork.us/Social-Services/311-Service-Requests-from-2020-to-Present/erm2-nwe9. Accessed: October 18, 2026.

Appendix: Data Documentation

A. Source, scope, and labeling

This appendix documents the saved NYC DOT Street Condition request dataset: 70,322 rows and 38 columns, with one request per row and creation dates in 2025. Counts were checked against repository commit `d4a15bc` on October 22, 2026. CSV files do not preserve dataframe dtypes; the types below describe interpretation. The companion GitHub data dictionary and category inventory provide the same definitions and machine-readable counts.

The source is NYC Open Data, dataset `erm2-nwe9`, *311 Service Requests from 2020 to Present*: `https://data.cityofnewyork.us/resource/erm2-nwe9.json`. Source field descriptions are available at `https://data.cityofnewyork.us/api/views/erm2-nwe9.json`. Python requests filtered the agency to DOT and the problem to Street Condition, using monthly creation-date windows from January 1, 2025 inclusive through January 1, 2026 exclusive. The automated helper requests ordered pages of 5,000 JSON records, preserves raw data, parses dates, and saves cleaned monthly CSVs.

The original 70,330 records became 70,324 after excluding six negative closure durations (February: 1; March: 2; July: 1; December: 2). The combined data had no duplicate request IDs. Two records with missing closure dates and Unspecified status were excluded from labeling. Recorded closure duration is the closure timestamp minus the creation timestamp, in hours. Dates are stored without timezone offsets, and the notebook uses timezone-naive subtraction; timezone and daylight-saving behavior should be confirmed before making precise elapsed-time claims.

Table 1: Rules used to construct the binary target.

Outcome rule	Label	Rows
Recorded closure duration from 0 to 48 hours inclusive	1	44,355
Recorded closure duration greater than 48 hours	0	22,492
Missing closure date with Pending or Open status	0	3,475
Missing closure date with Unspecified status	Excluded	2

The final target has 44,355 ones (63.1%) and 25,967 zeros (36.9%), with no missing labels. The 48-hour threshold is a team choice, not a verified city service standard or a measure of task difficulty. Closure means recorded administrative closure, not necessarily physical repair.

Labeling missing-closure Pending/Open requests as 0 assumes at least 48 hours of follow-up and sufficiently complete closure records. The 2025 requests were collected during the team's 2026 work, providing ample elapsed calendar time; record completeness remains an assumption. Missing-closure requests were retained, not all removed. Six Pending records have recorded closure times, so their labels follow the duration rule. There are 66,847 recorded durations, 3,475 missing durations, and 3,276 retained zero-hour durations. The live API changes, so later downloads need not reproduce this snapshot. Exact per-request retrieval times were not logged.

B. Column definitions

The saved file contains both possible predictors and audit/outcome fields. Roles describe an initial proposal, not a final fitted matrix. Candidate fields require a submission-time availability check; geovalidated values may have changed since submission. Derive means use calendar features. Leakage means exclude outcome information from predictors. Audit, Constant, Empty, Sparse, and Redundant are excluded initially. Target is the response, not a predictor. No ordered severity categories are defined. Null counts refer to blank CSV cells, not literal Unknown or Unspecified values.

Table 2: All 38 saved columns, semantic types, proposed roles, and missing counts.

Column / type / role	Definition and units	Missing
`unique_key` Identifier *Audit*	Unique service-request identifier. Keep as text for duplicate checks and traceability; exclude from predictors.	0

Column / type / role	Definition and units	Missing
`created_date` Timestamp *Derive*	Request creation date and time. Parse as a datetime; derive calendar features rather than use the raw timestamp.	0
`closed_date` Timestamp *Leakage*	Date and time the responding agency recorded closure. Used to calculate the outcome; exclude from predictors.	3,475
`agency` Nominal *Constant*	Responding agency acronym. DOT means Department of Transportation; constant after filtering.	0
`agency_name` Nominal *Constant*	Full responding agency name; Department of Transportation in every row.	0
`complaint_type` Nominal *Constant*	Broad problem category. Street Condition in every row; descriptor provides the varying problem detail.	0
`descriptor` Nominal *Candidate*	Specific street-condition problem, with 21 observed categories defined below. These labels are not an ordered severity scale.	0
`incident_zip` Nominal code *Candidate*	Geovalidated incident ZIP code. Preserve as text; numerical magnitude has no predictive interpretation. Codes are listed in the category inventory.	749
`intersection_street_1` Text *Audit*	First intersecting street from the geovalidated incident location. High-cardinality location text.	24,352
`intersection_street_2` Text *Audit*	Second intersecting street from the geovalidated incident location. High-cardinality location text.	24,331
`address_type` Nominal *Candidate*	Type of incident-location information, such as an address, intersection, or street segment. Six observed labels are defined below.	0
`city` Nominal *Audit*	City or locality name returned by geovalidation. Values are not necessarily borough names; partly redundant with other geography.	10,900
`facility_type` Nominal *Empty*	City facility type associated with a request, if available. Entirely blank in this snapshot.	70,322
`status` Nominal *Leakage*	Request status at collection, not necessarily status at submission. Closed, Pending, and Open are defined below. Used for labeling only.	0
`resolution_description` Text *Leakage*	Last action described by the responding agency, sometimes including next steps. Exclude because it can reveal post-submission handling.	15
`resolution_action_updated_date` Timestamp *Leakage*	Date and time the responding agency last updated the request. Post-submission information; exclude from predictors.	12
`community_board` Nominal code *Candidate*	Geovalidated community-board label, typically a board number plus borough. Exceptional labels are retained as source values and need validation.	0
`council_district` Nominal code *Candidate*	City Council district containing the request. Values 1 through 51 identify districts, not a ranking; saved CSV uses decimal-looking codes.	19,796
`police_precinct` Nominal code *Candidate*	NYPD precinct containing the request. Precinct numbers identify jurisdictions, not a ranking; Unspecified means not identified.	0

Column / type / role	Definition and units	Missing
`borough` Nominal *Candidate*	Incident borough, submitted and confirmed by geovalidation. Five NYC boroughs plus Unspecified.	0
`x_coordinate_state_plane` Continuous *Audit*	Geovalidated horizontal coordinate in the source state-plane system. The source field description does not specify units or CRS; verify before interpreting distances. Redundant with latitude/longitude.	19,598
`y_coordinate_state_plane` Continuous *Audit*	Geovalidated vertical coordinate in the source state-plane system. The source field description does not specify units or CRS; verify before interpreting distances. Redundant with latitude/longitude.	19,591
`open_data_channel_type` Nominal *Candidate*	How the request was submitted: PHONE, ONLINE, OTHER, or UNKNOWN in this snapshot.	0
`park_facility_name` Text *Constant*	Name of a Parks Department facility, if applicable. Every saved value is the placeholder Unspecified.	0
`park_borough` Nominal *Redundant*	Source park-related borough field. In this snapshot its values match borough row for row, so retain only borough for initial modeling.	0
`latitude` Continuous *Candidate*	Incident latitude in decimal degrees. Missing values must be handled within the training pipeline.	19,601
`longitude` Continuous *Candidate*	Incident longitude in decimal degrees. Missing values must be handled within the training pipeline.	19,601
`location` Structured text *Redundant*	Serialized geographic point containing longitude then latitude. Duplicates the separate coordinate fields; do not treat as a third independent location measurement.	19,601
`incident_address` Text *Audit*	Incident-address text supplied for the location. High-cardinality location field; retain for audit rather than the initial predictor matrix.	14,092
`street_name` Text *Audit*	Street name of the incident address supplied by the submitter. High-cardinality location text.	14,092
`cross_street_1` Text *Audit*	First cross street associated with the geovalidated incident location. High-cardinality location text.	14,511
`cross_street_2` Text *Audit*	Second cross street associated with the geovalidated incident location. High-cardinality location text.	14,494
`descriptor_2` Nominal *Sparse*	Third level of problem detail. Only Other appears, in 20 rows; the remaining 70,302 rows are blank. Exclude initially because it is nearly empty.	70,302
`location_type` Nominal *Candidate*	Type of location associated with the request, with six observed labels defined below. Frequently missing; optional candidate requiring missing-value handling.	38,771
`landmark` Text *Audit*	Landmark name associated with the incident location when available. High-cardinality location text.	48,935
`bbl` Identifier *Audit*	Borough-Block-Lot parcel identifier from geovalidation. Preserve as an identifier, not a continuous measurement; decimal suffixes are serialization artifacts.	43,894
`closure_hours` Continuous *Leakage*	Derived recorded duration in hours: (closed_date - created_date).total_seconds() / 3600. Missing when closure is unavailable; excludes negative durations. Never include in predictors.	3,475

Column / type / role	Definition and units	Missing
`closed_48h` Binary target *Target*	1 means recorded administrative closure within 48 hours inclusive. 0 means closure after 48 hours, or missing closure with Pending/Open status at collection. See target rules and follow-up assumption below.	0

C. Category meanings

The following explanations interpret source labels in plain language, not as official engineering severity definitions. Categories are nominal. Blank values mean missing information and are distinct from literal Unspecified or UNKNOWN labels. Counts and the complete geographic code inventory are also provided in `data/category_inventory.csv` in GitHub.

`descriptor`

Table 3: Observed categories for descriptor.

Value	Rows	Meaning
Blocked - Construction	4,245	Reported street obstruction associated with construction.
Cave-in	9,647	Reported collapse or sinking of the street surface.
Crash Cushion Defect	225	Reported defect in a crash cushion, an impact-absorbing road safety device.
Defacement	20	Reported defacement of a street-related surface or feature.
Defective Hardware	5,490	Reported defective street hardware; the label does not identify a specific component.
Depression Maintenance	31	Reported maintenance issue involving a depression in the street surface.
Dumpster - Construction Waste	584	Reported street condition involving a construction-waste dumpster.
Failed Street Repair	3,487	Reported problem with a previous street repair.
Guard Rail - Street	106	Reported street guard-rail condition.
Hummock	18	Reported raised bump or mound in the street surface.
Line/Marking - After Repaving	507	Reported road line or marking issue after repaving.
Line/Marking - Faded	386	Reported faded road line or marking.
Plate Condition - Anti-Skid	53	Reported issue with a street plate anti-skid surface.
Plate Condition - Noisy	2,927	Reported noise from a street plate.
Plate Condition - Open	65	Reported open street-plate condition; no engineering severity is inferred.
Plate Condition - Shifted	622	Reported street plate that has moved out of position.
Pothole	38,430	Reported hole in the street surface.
Rough, Pitted or Cracked Roads	3,121	Reported roughness, pitting, or cracking of the road surface.
Strip Paving	1	Reported condition classified by DOT as Strip Paving; the source label supplies no further detail.
Unsafe Worksite	66	Reported unsafe condition at a worksite.
Wear & Tear	291	Reported deterioration through wear and tear.

`address_type`

Table 4: Observed categories for address_type.

Value	Rows	Meaning
ADDRESS	27,748	Street address.

Value	Rows	Meaning
BLOCKFACE	19,639	Street segment or block face.
INTERSECTION	22,779	Intersection of streets.
PLACE	72	Source place-based label; exact distinction from PLACENAME is not documented in the schema.
PLACENAME	26	Source place-name label; exact distinction from PLACE is not documented in the schema.
UNRECOGNIZED	58	Location type not recognized by the source.

`status`

Table 5: Observed categories for status.

Value	Rows	Meaning
Closed	66,841	Agency-recorded closed status.
Open	8	Agency-recorded open status.
Pending	3,473	Agency-recorded pending status.

`borough`

Table 6: Observed categories for borough.

Value	Rows	Meaning
BRONX	6,955	NYC borough: Bronx.
BROOKLYN	18,375	NYC borough: Brooklyn.
MANHATTAN	12,386	NYC borough: Manhattan.
QUEENS	25,980	NYC borough: Queens.
STATEN ISLAND	6,494	NYC borough: Staten Island.
Unspecified	132	Borough not identified.

`open_data_channel_type`

Table 7: Observed categories for open_data_channel_type.

Value	Rows	Meaning
ONLINE	15,842	Submitted through an online channel.
OTHER	1	Other submission channel.
PHONE	13,191	Submitted by telephone.
UNKNOWN	41,288	Submission channel not known or not recorded.

`location_type`

Table 8: Observed categories for location_type.

Value	Rows	Meaning
Bike Lane	39	Bicycle lane.

Value	Rows	Meaning
Crosswalk	337	Pedestrian crossing.
School Safety Zone	14	School safety zone.
Speed Reducer	48	Traffic-calming speed reducer.
Street	31,006	Street location.
Vehicle Lane	107	Vehicle traffic lane.

agency

Table 9: Observed categories for agency.

Value	Rows	Meaning
DOT	70,322	Department of Transportation.

agency_name

Table 10: Observed categories for agency_name.

Value	Rows	Meaning
Department of Transportation	70,322	Full name of the responding agency.

complaint_type

Table 11: Observed categories for complaint_type.

Value	Rows	Meaning
Street Condition	70,322	Broad category of street-condition requests.

descriptor_2

Table 12: Observed categories for descriptor_2.

Value	Rows	Meaning
Other	20	Additional problem detail recorded as Other; no more specific detail provided.

closed_48h

Table 13: Observed categories for closed_48h.

Value	Rows	Meaning
0	25,967	Not recorded as closed within 48 hours under the labeling rules.
1	44,355	Recorded as closed within 48 hours inclusive.

park_facility_name

Table 14: Observed categories for park_facility_name.

Value	Rows	Meaning
Unspecified	70,322	No specific park facility identified.

Geographic categories

ZIP codes, community boards, council districts, precincts, and locality names are nominal location identifiers. Numbers identify geographic units; they do not define an ordinal scale. Community-board labels may contain exceptional or nonstandard values that need validation. Council codes such as 1.0 are serialization of district IDs, not measured values. Each listed locality name denotes the source geovalidated locality, which is not necessarily a borough name. Park borough has the same six categories and row-level values as borough in this snapshot, so it is redundant. The lists below enumerate all nonblank geographic categories; the GitHub inventory supplies individual counts and meanings.

incident_zip 10000; 10001; 10002; 10003; 10004; 10005; 10006; 10007; 10009; 10010; 10011; 10012; 10013; 10014; 10016; 10017; 10018; 10019; 10020; 10021; 10022; 10023; 10024; 10025; 10026; 10027; 10028; 10029; 10030; 10031; 10032; 10033; 10034; 10035; 10036; 10037; 10038; 10039; 10040; 10041; 10045; 10048; 10065; 10069; 10075; 10103; 10105; 10110; 10111; 10112; 10118; 10119; 10122; 10123; 10128; 10151; 10153; 10158; 10168; 10169; 10170; 10173; 10174; 10176; 10179; 10271; 10278; 10279; 10280; 10281; 10282; 10301; 10302; 10303; 10304; 10305; 10306; 10307; 10308; 10309; 10310; 10312; 10314; 10451; 10452; 10453; 10454; 10455; 10456; 10457; 10458; 10459; 10460; 10461; 10462; 10463; 10464; 10465; 10466; 10467; 10468; 10469; 10470; 10471; 10472; 10473; 10474; 10475; 11001; 11004; 11005; 11040; 11101; 11102; 11103; 11104; 11105; 11106; 11109; 11201; 11203; 11204; 11205; 11206; 11207; 11208; 11209; 11210; 11211; 11212; 11213; 11214; 11215; 11216; 11217; 11218; 11219; 11220; 11221; 11222; 11223; 11224; 11225; 11226; 11228; 11229; 11230; 11231; 11232; 11233; 11234; 11235; 11236; 11237; 11238; 11239; 11242; 11249; 11354; 11355; 11356; 11357; 11358; 11359; 11360; 11361; 11362; 11363; 11364; 11365; 11366; 11367; 11368; 11369; 11370; 11372; 11373; 11374; 11375; 11377; 11378; 11379; 11385; 11411; 11412; 11413; 11414; 11415; 11416; 11417; 11418; 11419; 11420; 11421; 11422; 11423; 11426; 11427; 11428; 11429; 11430; 11432; 11433; 11434; 11435; 11436; 11691; 11692; 11693; 11694; 11697.

city ARVERNE; ASTORIA; BAYSIDE; BELLEROSE; BREEZY POINT; BRONX; BROOKLYN; CAMBRIA HEIGHTS; COLLEGE POINT; CORONA; EAST ELMHURST; ELMHURST; FAR ROCKAWAY; FLORAL PARK; FLUSHING; FOREST HILLS; FRESH MEADOWS; GLEN OAKS; HOLLIS; HOWARD BEACH; JACKSON HEIGHTS; JAMAICA; KEW GARDENS; LITTLE NECK; LONG ISLAND CITY; MANHATTAN; MASPETH; MIDDLE VILLAGE; NEW HYDE PARK; NEW YORK; OAKLAND GARDENS; OZONE PARK; QUEENS; QUEENS VILLAGE; REGO PARK; RICHMOND HILL; RIDGEWOOD; ROSEDALE; SAINT ALBANS; SOUTH OZONE PARK; SOUTH RICHMOND HILL; SPRINGFIELD GARDENS; STATEN ISLAND; SUNNYSIDE; WHITESTONE; WOODHAVEN; WOODSIDE.

community_board 0 Unspecified; 01 BRONX; 01 BROOKLYN; 01 MANHATTAN; 01 QUEENS; 01 STATEN ISLAND; 02 BRONX; 02 BROOKLYN; 02 MANHATTAN; 02 QUEENS; 02 STATEN ISLAND; 03 BRONX; 03 BROOKLYN; 03 MANHATTAN; 03 QUEENS; 03 STATEN ISLAND; 04 BRONX; 04 BROOKLYN; 04 MANHATTAN; 04 QUEENS; 05 BRONX; 05 BROOKLYN; 05 MANHATTAN; 05 QUEENS; 06 BRONX; 06 BROOKLYN; 06 MANHATTAN; 06 QUEENS; 07 BRONX; 07 BROOKLYN; 07 MANHATTAN; 07 QUEENS; 08 BRONX; 08 BROOKLYN; 08 MANHATTAN; 08 QUEENS; 09 BRONX; 09 BROOKLYN; 09 MANHATTAN; 09 QUEENS; 10 BRONX; 10 BROOKLYN; 10 MANHATTAN; 10 QUEENS; 11 BRONX; 11 BROOKLYN; 11 MANHATTAN; 11 QUEENS; 12 BRONX; 12 BROOKLYN; 12 MANHATTAN; 12 QUEENS; 13 BROOKLYN; 13 QUEENS; 14 BROOKLYN; 14 QUEENS; 15 BROOKLYN; 16 BROOKLYN; 17 BROOKLYN; 18 BROOKLYN; 26 BRONX; 27 BRONX; 28 BRONX; 55 BROOKLYN; 56 BROOKLYN; 64 MANHATTAN; 80 QUEENS; 81 QUEENS; 82 QUEENS; 83 QUEENS; 84 QUEENS; 95 STATEN ISLAND; Unspecified BRONX; Unspecified BROOKLYN; Unspecified MANHATTAN; Unspecified QUEENS; Unspecified STATEN ISLAND.

council_district 1.0; 10.0; 11.0; 12.0; 13.0; 14.0; 15.0; 16.0; 17.0; 18.0; 19.0; 2.0; 20.0; 21.0; 22.0; 23.0; 24.0; 25.0; 26.0; 27.0; 28.0; 29.0; 3.0; 30.0; 31.0; 32.0; 33.0; 34.0; 35.0; 36.0; 37.0; 38.0; 39.0; 4.0; 40.0; 41.0; 42.0; 43.0; 44.0; 45.0; 46.0; 47.0; 48.0; 49.0; 5.0; 50.0; 51.0; 6.0; 7.0; 8.0; 9.0.

`police_precinct` Precinct 1; Precinct 10; Precinct 100; Precinct 101; Precinct 102; Precinct 103; Precinct 104; Precinct 105; Precinct 106; Precinct 107; Precinct 108; Precinct 109; Precinct 110; Precinct 111; Precinct 112; Precinct 113; Precinct 114; Precinct 115; Precinct 120; Precinct 121; Precinct 122; Precinct 123; Precinct 13; Precinct 14; Precinct 17; Precinct 18; Precinct 19; Precinct 20; Precinct 22; Precinct 23; Precinct 24; Precinct 25; Precinct 26; Precinct 28; Precinct 30; Precinct 32; Precinct 33; Precinct 34; Precinct 40; Precinct 41; Precinct 42; Precinct 43; Precinct 44; Precinct 45; Precinct 46; Precinct 47; Precinct 48; Precinct 49; Precinct 5; Precinct 50; Precinct 52; Precinct 6; Precinct 60; Precinct 61; Precinct 62; Precinct 63; Precinct 66; Precinct 67; Precinct 68; Precinct 69; Precinct 7; Precinct 70; Precinct 71; Precinct 72; Precinct 73; Precinct 75; Precinct 76; Precinct 77; Precinct 78; Precinct 79; Precinct 81; Precinct 83; Precinct 84; Precinct 88; Precinct 9; Precinct 90; Precinct 94; Unspecified.

D. Derived features and modeling boundary

The committed EDA notebook creates hour (0–23), month (1–12, January–December), day_of_week (Monday–Sunday), and location_missing (True when latitude is absent, False otherwise) in memory. They are not additional columns in the saved 38-column CSV. Hour, month, and weekday have cyclical calendar interpretations rather than a street-condition severity order. The notebook replaces borough Unspecified with missing in its working copy. Coordinates are missing in 19,601 rows (27.9%).

Weekend status and historical complaint-volume features discussed in the report remain planned; they are not created in the committed EDA notebook. If implemented, weekend denotes Saturday or Sunday. Historical counts must use only requests strictly earlier than the current request, specify the grouping and time window, and provide earlier history or an insufficient-history rule for the first window.

Exclude closed_date, closure_hours, status, resolution_description, and resolution_action_updated_date from predictors because they can reveal the outcome. Use closed_48h only as the response. Retain unique_key for audit, not prediction. Other initial exclusions are given in Table 2. Missing-value handling and encoding must be learned within the training pipeline. Unique IDs do not imply independence: requests can be related by place, time, or underlying incident. No final predictor matrix, validation split, trained model, or predictive performance is established by this documentation.